Yu-Chu Yu

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Education

M.S. in Computer Science and Information Engineering
2020/09 - 2023/01

National Taiwan University

- **GPA:** 4.1 / 4.3
- **Adivisor: Hsuan-Tien Lin**. Computational Learning Lab (CLLab)
- **Honor:** *Master Thesis Award*. Taiwanese Association for Artificial Intelligence 2023
- **Master Thesis:** Semi-Supervised Domain Adaptation with Source Label Adaptation (CVPR'23)

B.S. in Computer Science and Information Engineering
2016/09 - 2020/06

National Taiwan University

Publication

[1] **Yu-Chu Yu**, Chieh Hubert Lin, Hsin-Ying Lee, Chaoyang Wang, Yu-Chiang Frank Wang, Ming-Hsuan Yang. "Towards Affordance-Aware Articulation Synthesis for Rigged Objects". *arXiv preprint.* January 2025. [Paper Link].

[2] Jr-Jen Chen, Yu-Chien Liao, Hsi-Che Lin, **Yu-Chu Yu**, Yen-Chun Chen, Yu-Chiang Frank Wang. "ReXTime: A Benchmark Suite for Reasoning-Across-Time in Videos". *Neural Information Processing Systems **(NeurIPS)**, Datasets and Benchmarks Track.* December 2024. [Paper Link].

[3] **Yu-Chu Yu**, Chi-Pin Huang, Jr-Jen Chen, Kai-Po Chang, Yung-Hsuan Lai, Fu-En Yang, Yu-Chiang Frank Wang. "Select and Distill: Selective Dual-Teacher Knowledge Transfer for Continual Learning on Vision-Language Models". *European Conference on Computer Vision **(ECCV)**.* September 2024. [Paper Link].

[4] **Yu-Chu Yu**, Hsuan-Tien Lin. "Semi-Supervised Domain Adaptation with Source Label Adaptation". *IEEE/CVF Conference on Computer Vision and Pattern Recognition **(CVPR)**.* June 2023. [Paper Link].

Research Experience

Visiting Student Researcher
2024/04 - Present

University of California, Merced. Advisor: Prof. Ming-Hsuan Yang

- Propose A3Syn [1], a novel framework to synthesize affordance-aware 3D articulation for open-domain rigged objects.

Research Assistant
2023/09 - Present

National Taiwan University. Advisor: Prof. Yu-Chiang Frank Wang

- Research on Continual Learning for Vision-Language Models, develop a selective knowledge distillation algorithm [3] to preserve both previous knowledge and zero-shot transferability for VLMs. (ECCV'24)

Graduate Student Researcher
2020/09 - 2023/01

National Taiwan University. Advisor: Prof. Hsuan-Tien Lin

- Research on Semi-Supervised Domain Adaptation.
- Address Domain Adaptation as a Noisy Label Learning problem. Develop a dynamic algorithm SLA [4] to learn from the noisy source labels and **outperform state-of-the-art algorithms**. (CVPR'23)

Work Experience

Machine Learning Engineer

2022/11 - 2023/09

Dcard Taiwan LTD. Manager: Ting-Hsuan Chao

- Design a recommendation system for a Taiwanese forum platform, serving over 15 million users with more than 10,000 posts per day.
- Develop an interactive API to recommend posts to users on-the-fly (< 1.8 sec. for 95% requests) according to their previous behaviours.

Research Development Intern

2021/12 - 2022/06

Microsoft @ Bing Maps & Geospatial Team. Manager: Jason Hong

- Develop an asynchronous web crawler system to automatically evaluate the performance of reverse geocoding services provided by google, bing and tomtom.

Teaching

Teaching Assistant

2020 Fall, 2021 Fall

CSIE 5043: Machine Learning. Instructor: Prof. Hsuan-Tien Lin

- Help introduce the foundational concepts of machine learning in 600-person class.
- Design assignments, and projects, and hold office hour for answering questions.

Teaching Assistant

2021 Spring

CSIE 1212: Data Structures and Algorithms. Instructor: Prof. Hsuan-Tien Lin & Prof. Hsin-Mu Tsai

- Help deal with administrative issues to better maintain a 400-person class.
- Provide office hour for debugging, answering questions.

Professional Service

Served as a reviewer for:

- **Machine Learning**: *NeurIPS'24*
- **Computer Vision**: *CVPR'25*

Award

Master Thesis Award

2023

Taiwanese Association for Artificial Intelligence